

A Three-Channel Floating-Electrode Capacitance Readout Board Based on the PCAP04 Capacitance-to-Digital Converter

Design Description, Independent Verification and Review
of Hardware Revision v1.5

Jana Baker

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Abstract

This document describes and verifies a compact acquisition board for three cable-connected capacitive sensors (nominally 64–256 pF each) used for capacitive level and position sensing. The board combines a ScioSense PCAP04 capacitance-to-digital converter (CDC), operated in *floating* mode with full internal and external parasitic compensation, with an STM32G0B1 microcontroller that provides the SPI host, a USB 2.0 full-speed CDC link and an explicit power-gating and tri-state interlock for the analogue section. The six sensor electrodes enter through edge-mounted SMA connectors; supply rails are derived from USB or an external 5 V input through separate digital and ultra-low-noise analogue regulators. The verification reported here was performed independently of the CAD tool: the schematic and board files (KiCad 10) were parsed programmatically, the resulting pin-net map was cross-checked against the manufacturers’ datasheets for all seven integrated circuits, and the copper geometry was analysed for signal-integrity and manufacturability. All 109 IC pin assignments were found to be correct and the board passes native ERC/DRC with zero violations. Eleven observations are reported, none of which prevents fabrication of the bare board; the most significant concern is the routing of signal tracks through the internal ground-plane layer, which contradicts the design’s own layout requirement. Recommended corrections for a v1.6 revision and a bring-up procedure are given.

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1 Introduction

Capacitive sensing offers a simple, drift-tolerant way of measuring the level of a dielectric liquid or the relative position of two conductors: the capacitance between a pair of electrodes changes with the fraction of the gap filled by the medium or with the electrode overlap. In instrumentation the electrodes are frequently remote from the electronics and are reached through coaxial cables, so that the readout must tolerate a cable capacitance to ground that is comparable to, or larger than, the quantity being measured.

The PCAP04 [1] is a single-chip capacitance-to-digital converter that measures capacitance ratiometrically by timing the discharge of an RC network. It supports up to six grounded or three floating capacitors, provides on-chip compensation of internal and external parasitic capacitance, and reports a resolution in the attofarad range at low data rates. These properties make it a natural choice for a small, cable-driven, multi-channel level sensor front end.

This document has two purposes. First, it gives a complete engineering description of the hardware revision v1.5 of the *PCAP04 Acquisition 3CH* board (100 × 100 mm, four copper layers), including the measurement principle, the circuit design rationale, and the printed-circuit-board (PCB) layout. Second, it reports an independent correctness review of that design. The review was deliberately carried out outside the CAD environment: the KiCad schematic and board files were parsed with purpose-written scripts to recover the pin-net connectivity and copper geometry, and every integrated-circuit pin was compared against the manufacturer’s datasheet. Design rules, decoupling, power sequencing, the analogue front end and the layout were then assessed against the datasheet requirements and against the design intent stated in the schematic notes. Section 2 summarises the measurement principle, Sections 3–5 describe the design, Section 6 the verification method and results, and Section 7 the findings and the recommendations for the next revision.

2 Measurement principle

2.1 Ratiometric discharge-time measurement

The PCAP04 charges the capacitor under test and then discharges it through an on-chip resistor R while a time-to-digital converter measures the interval until the port voltage crosses a fixed comparator threshold. For a capacitor C the discharge is exponential, so that the measured interval is

$$\tau = k R C, \quad k = \ln\left(\frac{V_0}{V_{th}}\right), \quad (1)$$

where V_0 is the pre-charge voltage (the 1.8 V core supply) and V_{th} the comparator threshold. The same resistor is used for a reference capacitor C_{ref} , so that the unknown constants k and R cancel in the ratio delivered by the on-chip DSP:

$$\frac{\tau_N}{\tau_{ref}} = \frac{C_N}{C_{ref}}. \quad (2)$$

Figure 1 illustrates the timing. The internal discharge resistor can be set to 10 kΩ, 30 kΩ, 90 kΩ or 180 kΩ separately for ports PC0–PC3 (with the internal reference PC6) and for ports PC4–PC5 [1]. With the default 30 kΩ and the largest sensor value of 256 pF the time constant is $RC \approx 7.7 \mu\text{s}$; including 150 pF of cable capacitance at the discharging node it rises to about 12 μs, well within the discharge-time range of the converter.

2.2 Floating sensors and parasitic compensation

Two connection schemes are available. In *grounded* mode one electrode of each sensor is tied to ground and up to six sensors share the reference PC0; the full cable capacitance of the active

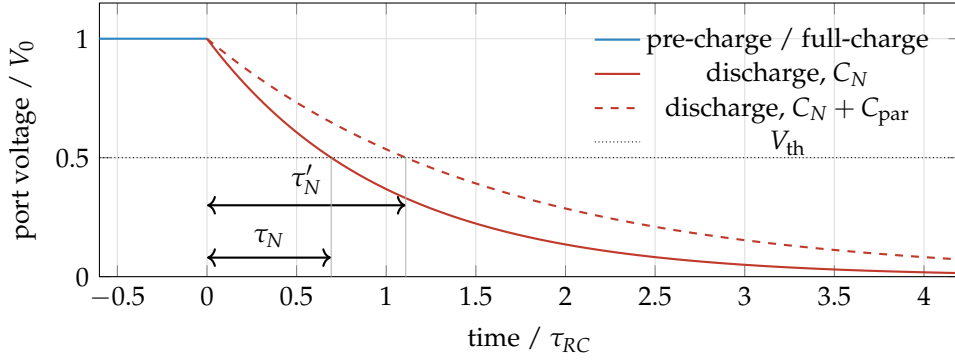


Figure 1: Discharge-time principle of the PCAP04. The interval from the start of the discharge to the comparator threshold crossing is proportional to RC . A parasitic capacitance to ground at the discharging node lengthens the interval (τ_N'); in floating mode this contribution is measured separately and removed (Section 2.2).

conductor then adds directly to the measured value. In *floating* mode each sensor is connected between two ports (PC0/PC1, PC2/PC3, PC4/PC5); only one electrode is charged while the other is switched to ground, and the parasitic capacitances of both conductors to ground can be separated from the sensor value [1]. This is the scheme adopted here (Figure 2). For each floating capacitor the converter performs three measurements: (i) electrode A charged, electrode B grounded, giving $C_{AB} + C_{pA} + C_{int}$; (ii) the mirror image, giving $C_{AB} + C_{pB} + C_{int}$; and (iii) both electrodes charged to the same potential, so that no field exists across the sensor and only $C_{pA} + C_{pB} + C_{int}$ is seen. A fourth measurement with all ports open (*internal compensation*) yields C_{int} together with the comparator delay. From these the DSP firmware solves for C_{AB} alone. The datasheet recommends grounding the cable shields at the PCB end, which is exactly the topology used on this board (Section 4.1).

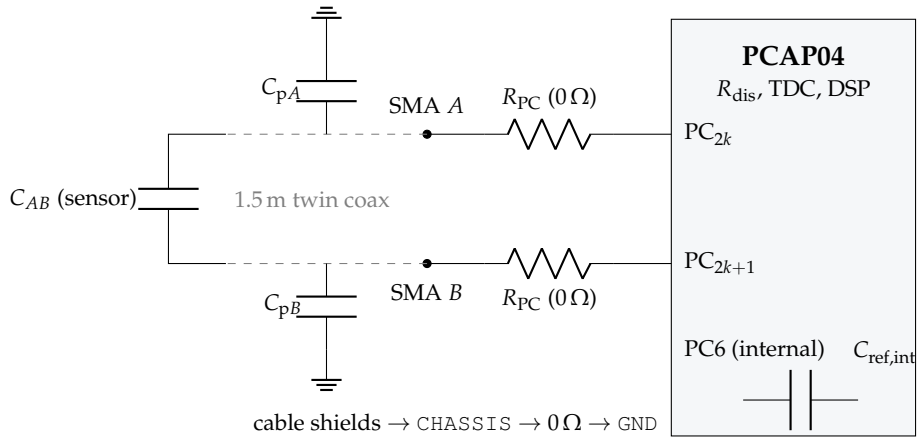


Figure 2: Floating connection of one sensor channel $k \in \{0, 1, 2\}$. The electrode pair is measured between two CDC ports; the coaxial shields are returned to the board ground at a single point. C_{pA} and C_{pB} (cable plus PCB trace) are removed by the external compensation sequence; the ratio is formed against the programmable internal reference capacitor on the internal port PC6.

2.3 Reference capacitor and dynamic range

Because all three port pairs are occupied by sensors, the design relies on the internal reference capacitor (register bit C_REF_INT), whose value is programmable as $C_{ref,int} \approx 0.959 \text{ pF} \cdot N + 3.23 \text{ pF}$ for $N = 0 \dots 31$, i.e. up to about 33 pF [1]. With the maximum setting the expected ratios lie between $64/33 \approx 1.9$ and $256/33 \approx 7.8$, which fits the 5-bit integer part of the result registers. Two consequences should be noted. First, the absolute accuracy of the readout is bounded by the

tolerance and temperature drift of the on-chip reference, so a gain calibration against a known external capacitor is required for absolute measurements; the board provides a footprint (CTEST1, 100 pF C0G 0.1 %, not fitted) across PC4/PC5 for that purpose. Second, if the highest long-term stability is required, one port pair (PC0/PC1) can instead carry an external reference capacitor ($C_{REF_INT}=0$), at the cost of reducing the sensor count from three to two; the hardware supports both options without modification.

2.4 Resolution

The datasheet characterises the converter with a 10 pF base capacitance: in floating mode with full compensation the RMS noise is 8 aF at 2.5 Hz, 56 aF at 100 Hz and 156 aF at 1 kHz [1]. Since the discharge-time jitter is, to first order, a fixed fraction of the measured interval, the noise scales roughly with the total capacitance at the port; for the 64–256 pF sensors plus cable used here, a relative resolution in the 10^{-5} range at data rates of a few hertz can therefore be expected, corresponding to a few femtofarads. The dominant systematic effects are the reference drift discussed above and the temperature coefficient of the cable dielectric; the design keeps the electronics at room temperature and compensates the cable through the three-measurement sequence.

3 System architecture

Figure 3 shows the organisation of the board. Three functional regions are separated on the PCB and in the schematic: the sensor/chassis region with the six SMA inputs and the CDC, the real-time control region with the microcontroller and the SPI buffer, and the USB/power region. Table 1 lists the active components.

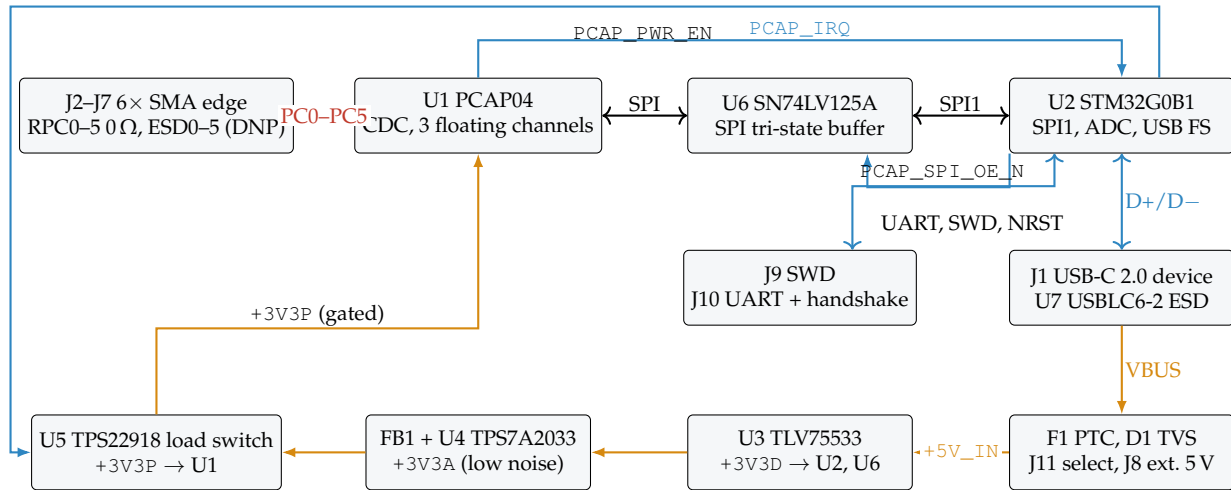


Figure 3: Block diagram of the acquisition board. Red: analogue sense path; blue: digital control; orange: power. The CDC supply +3V3P is gated by the microcontroller, and the SPI lines are tri-stated while the CDC is unpowered.

Signal flow. Each SMA centre conductor reaches a CDC port through a 0 Ω series element (RPC0–RPC5) that reserves the option of a small series resistor, and has a not-fitted ESD-diode position to CHASSIS. The PCAP04 is read over a four-wire SPI bus. The bus does not connect to the microcontroller directly but through the quad buffer U6, whose output-enable lines are driven by the microcontroller and pulled inactive by default; this prevents the buffer outputs from back-feeding the CDC through its I/O protection diodes while its supply is switched off. The interrupt output of the CDC (PCAP_IRQ) goes to an EXTI-capable microcontroller pin through a

Table 1: Active components of revision v1.5.

Ref.	Part	Function	Package
U1	ScioSense PCAP04-AQFM-24	Capacitance-to-digital converter, SPI mode	QFN-24 4×4
U2	ST STM32G0B1CBT6	Cortex-M0+ host: SPI master, ADC, USB 2.0 FS device, UART	LQFP-48
U3	TI TLV75533PDBVR	3.3 V/500 mA LDO, digital rail +3V3D	SOT-23-5
U4	TI TPS7A2033PDBVR	3.3 V/300 mA ultra-low-noise LDO, analogue rail +3V3A	SOT-23-5
U5	TI TPS22918DBVR	Load switch with slew control and quick output discharge, gated rail +3V3P	SOT-23-6
U6	TI SN74LV125APWR	Quad tri-state buffer isolating the SPI bus	TSSOP-14
U7	ST USBLC6-2SC6	USB ESD protection	SOT-23-6
J2–J7	BAT Wireless BWSMA-KE-P001 (LCSC C496550)	Edge-mount SMA, 1.6 mm PCB	edge clamp
J1	HRO TYPE-C-31-M-12	USB-C 2.0 receptacle, 16 pin	SMD/THT

1 k Ω series resistor. Two handshake lines (HOST_READY, MEAS_BUSY) and a UART are brought to header J10 for an external host; USB CDC over J1 is the primary data link.

Power domains. Three 3.3 V rails are used deliberately. +3V3D (U3) supplies the microcontroller and the SPI buffer; +3V3A (U4, preceded by ferrite bead FB1) is the low-noise analogue rail; +3V3P is +3V3A passed through the load switch U5 and supplies only the CDC. The microcontroller can thus power-cycle the CDC under firmware control (e.g. to recover from a latched state or to enforce a clean configuration sequence), and can verify through a resistive divider on ADC input PA0 (PCAP_PWR_SENSE) that the rail has actually collapsed before re-enabling it.

4 Circuit design

4.1 Sensor front end

Figure 4 shows one of the three identical channels together with the shared chassis network. The SMA outer conductors of all six connectors and the USB receptacle shell form the CHASSIS net, which is bonded to signal ground at exactly one point through the 0 Ω link RCH0. Alternative positions for a 1 M Ω bleed resistor (RCH1) and a 1 nF capacitor (CCH0) are provided but not fitted, so that the single-point bond can be replaced by a soft ground if the installation requires it. The schematic note explicitly prohibits a second chassis-to-ground connection. This arrangement satisfies the PCAP04 recommendation that cable shields be grounded at the PCB end [1], keeps the shield return currents away from the CDC ground pins, and defines the potential of the parasitic capacitances that are removed by the compensation sequence.

The series elements RPC n are fitted as 0 Ω . Any series resistance adds directly to the on-chip discharge resistor in Eq. (1); for the 30 k Ω setting a 100 Ω part would introduce a 0.3 % gain change that is common to sensor and (internal) reference only if the reference path has the same resistance, which it does not. Keeping them at 0 Ω is therefore the correct choice for accuracy; the footprints remain useful for a ferrite bead or a fusible link should EMI or fault tolerance require it. The ESD positions ESD0–ESD5 (SOD-882, to CHASSIS) are not fitted on the first build because

any clamp diode adds a voltage-dependent capacitance and leakage in parallel with the sensor; they can be populated with a low-capacitance (< 0.5 pF) device if field experience shows a need.

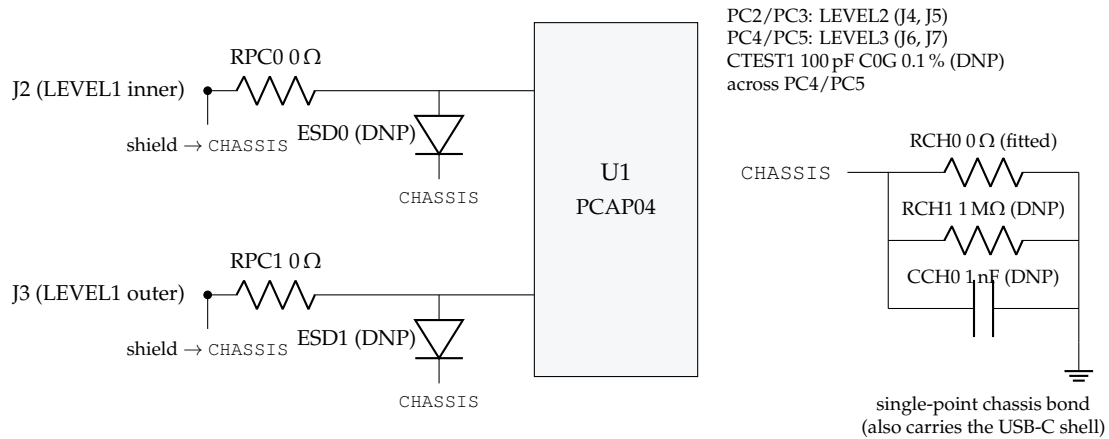


Figure 4: Sensor front end for channel 1 (PC0/PC1) and the shared chassis network. Channels 2 and 3 are identical.

4.2 CDC and its supplies

The PCAP04 is powered from the gated rail +3V3P on both VDD33 pins (4 and 14). The datasheet requires a buffer capacitor of at least 10 μ F on VDD33 and at least 4.7 μ F on the 1.8 V core-regulator output VDD18 [1]. The board fits 100 nF directly at each VDD33 pin (C1, C2, within 2 mm) plus two 10 μ F bulk capacitors (C3, C7), and 4.7 μ F (C4) at VDD18 with an optional 100 nF position (C5). VDD18 is a regulator *output*; the schematic warns that it must never be connected to +3V3P, and the netlist confirms that it is a private net (VDD18_LOCAL) with only the two capacitors attached. All four ground pins including the exposed pad are on GND. The interface select pin IIC_EN (13) is tied hard to ground, selecting the SPI interface; the datasheet states that this pin must not float. The temperature-measurement ports PT0REF, PT1 and PTOUT and the auxiliary/guard port PCAUX are left open, as permitted when the resistance-to-digital function and guarding are unused (guarding is in any case not available in floating mode). The general-purpose ports PG2–PG4 are unconnected; PG5 carries the interrupt output once PG5_INTN_EN is set. A 100 k Ω pull-up (RCS_PU1) referenced to +3V3P holds the chip-select line inactive whenever the CDC is powered and the buffer is tri-stated.

4.3 SPI isolation and CDC power gating

Figure 5 shows the interlock between the two supply domains. The three host-to-CDC lines (SCK, MOSI, CS) pass through buffers 1–3 of U6, the CDC-to-host line MISO through buffer 4; all four output-enable pins are on PCAP_SPI_OE_N, pulled to +3V3D by ROEPU1 so that the bus is tri-stated at reset and whenever the microcontroller pin is high-impedance. Series resistors of 33 Ω on the three driven lines damp reflections and limit current in the event of contention. The SN74LV125A's I_{off} specification allows its outputs to sit on an unpowered node [6], so the buffer outputs may legitimately face the unpowered CDC.

The load switch U5 is controlled by PCAP_PWR_EN with a 100 k Ω pull-down (RENPD1), i.e. the CDC is *off* until firmware asserts the enable; the ON-pin threshold of 1 V is easily met by the 3.3 V GPIO [5]. The 1 nF slew capacitor CCT1 gives a specified rise time of about 1.7 ms at 3.3 V, so that the inrush into the 20 μ F of CDC decoupling is limited to roughly 40 mA. The quick-output-discharge pin QOD is returned to +3V3P through RQOD1 (470 Ω); with the internal pull-down of about 25 Ω the switched-off rail decays with $\tau \approx 495 \Omega \times 20 \mu\text{F} \approx 10$ ms and reaches the 50 mV threshold defined in the schematic in about 40 ms. The rail is monitored through RPS1/RPS2

(330 k Ω /100 k Ω , 0.77 V at full rail) with a 10 nF C0G filter capacitor; the 77 k Ω source impedance requires a long ADC sampling time (or reliance on the filter capacitor as a charge reservoir), which is a firmware setting. The CDC interrupt line has a 10 k Ω pull-down so that it reads as a defined level while the CDC is unpowered.

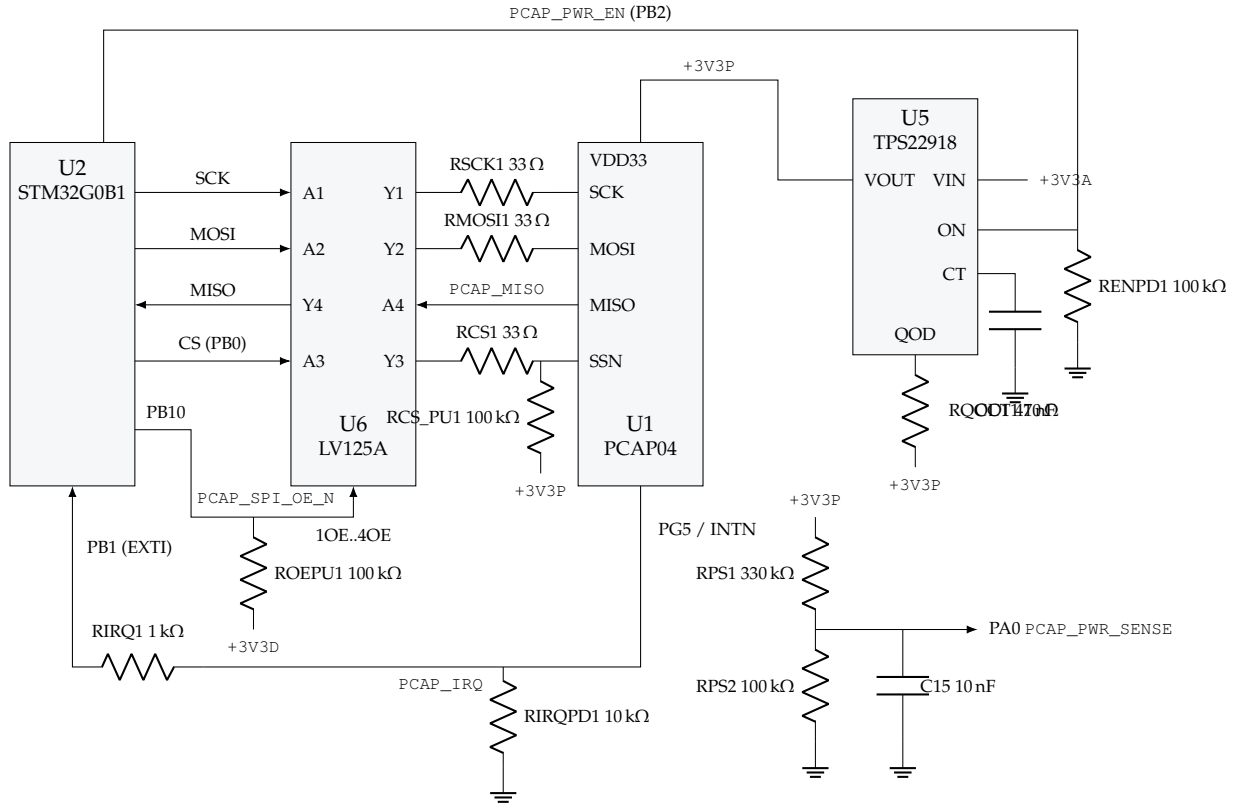


Figure 5: SPI isolation and power gating of the CDC. The buffer outputs are enabled only while the CDC rail is up; the chip-select pull-up is referenced to the gated rail; the rail is switched with a controlled slew and actively discharged, and its state is read back by the ADC.

4.4 Microcontroller

The STM32G0B1CBT6 (LQFP-48, general-purpose variant) is chosen for its crystal-less USB full-speed device peripheral, which uses the internal 48 MHz oscillator trimmed by the clock-recovery system from the USB start-of-frame packets [2]. Accordingly no crystal is fitted: PF0/PF1 and PC14/PC15 are left open. All supply pins (VDD/VDDA, VREF+, VBAT) are tied to +3V3D, which the datasheet permits (VREF+ may equal VDDA; VBAT may be tied to VDD when no backup source exists). Decoupling is 100 nF on VDD/VDDA and VREF+ and 4.7 μ F on VBAT (C8, C9, C10). NRST carries the recommended 100 nF capacitor, an additional 10 k Ω pull-up in parallel with the internal 25–55 k Ω pull-up, and the reset push-button SW1; the line is also available on the SWD header J9. SWCLK is shared with BOOT0 on PA14; no capacitor is placed on that pin so that debugging is unaffected, and boot-mode selection is left to the nBOOT_SEL option bit. Because a blank device automatically enters the system bootloader (flash empty check), first programming is possible over USB DFU without a debugger.

The pin usage is summarised in Table 2. SPI1 uses PA5/PA6/PA7 with PB0 as software chip-select, the CDC interrupt enters on PB1, the two control outputs are PB2 (power enable) and PB10 (buffer enable), PA0 is the ADC rail monitor, USART2 on PA2/PA3 and two GPIO handshake lines go to J10, and USB uses PA11/PA12. The remaining 25 I/O pins are unconnected and must be configured as pulled inputs or analogue in firmware.

Table 2: Microcontroller pin allocation (verified against the LQFP-48 pinout of [2]).

Pin	Port	Net	Role
4, 5, 6	VBAT, VREF+, VDD/VDDA	+3V3D	supply
7	VSS/VSSA	GND	ground
10	PF2-NRST	NRST	reset (RC, button, SWD)
11	PA0	PCAP_PWR_SENSE	ADC, CDC rail monitor
13, 14	PA2, PA3	UART_TX, UART_RX	USART2 to J10
16, 17, 18	PA5, PA6, PA7	SPI1 SCK, MISO, MOSI	to/from U6
19	PB0	PCAP_CS	SPI chip-select (GPIO)
20	PB1	PCAP_IRQ (via 1 k Ω)	EXTI from CDC
21	PB2	PCAP_PWR_EN	load-switch ON
22	PB10	PCAP_SPI_OE_N	buffer output enable (active low)
23, 24	PB11, PB12	HOST_READY, MEAS_BUSY	handshake to J10
33, 34	PA11, PA12	USB_DM, USB_DP	USB FS device
35, 36	PA13, PA14	SWDIO, SWCLK_BOOT0	SWD / BOOT0

4.5 USB interface and power input

The USB-C receptacle is wired as a legacy USB 2.0 upstream-facing port (Figure 6): CC1 and CC2 each carry a 5.1 k Ω pull-down (RCC1, RCC2) so that a Type-C source or a C-to-A cable supplies VBUS; both D+ and both D– contacts are paired; the SBU contacts are unconnected; the four VBUS contacts are joined. VBUS passes through a 500 mA PTC fuse F1 and is clamped by the SMF5.0A TVS D1 to form +5V_USB, which also powers the VBUS pin of the ESD array U7. The data lines pass through U7 (USBLC6-2SC6), whose two I/O pins per line are used as the “in” and “out” sides so that the clamp is in the signal path close to the connector. The connector shell is on CHASSIS. The 5 V source is selected with jumper J11 between +5V_USB and the external low-noise input J8 (+5V_EXT); the two sources are never paralleled. The selected +5V_IN feeds the digital LDO directly and the analogue LDO through the ferrite bead FB1 (600 Ω at 100 MHz); each LDO has its enable pin tied to its input and input/output capacitors that exceed the datasheet minima of 0.47 μ F effective (TLV755P) and 1 μ F (TPS7A20) [3, 4]. The TPS7A20 contributes 7 μ V_{RMS} of output noise and more than 60 dB of PSRR at 100 kHz, so that switching noise from a USB host supply is strongly attenuated before it reaches the CDC.

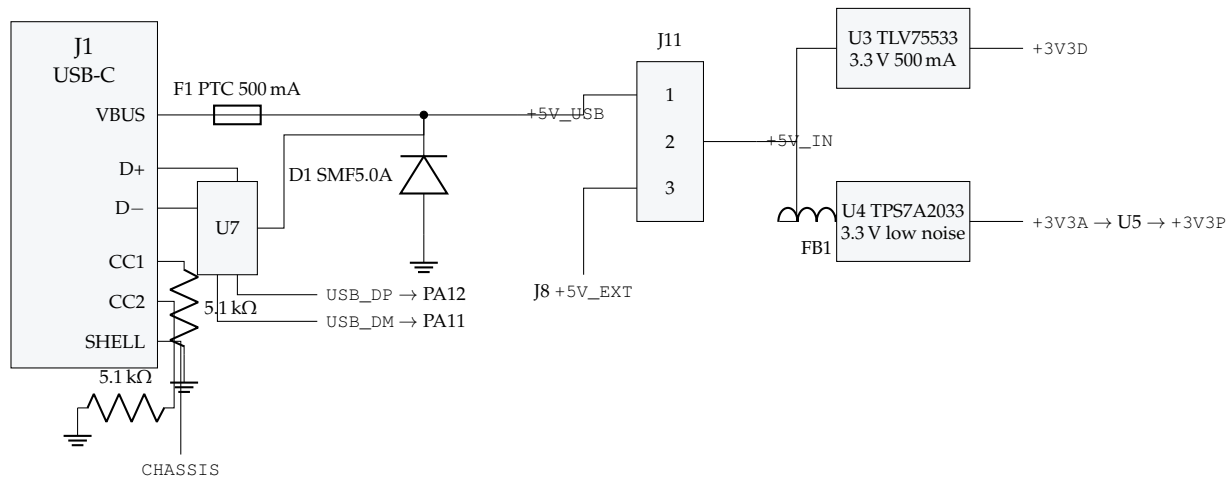


Figure 6: USB-C input, protection, source selection and regulator tree. Decoupling capacitors are omitted for clarity (values in Appendix A).

5 Printed-circuit-board layout

5.1 Stack-up, outline and connectors

The board is a 100×100 mm, 1.6 mm four-layer FR-4 design with the layer functions listed in Table 3. No explicit stack-up is stored in the board file, so the fabricator’s default four-layer build applies; for the estimates below the common JLC04161H-7628 construction (0.21 mm prepreg between the outer layers and the adjacent planes, $\epsilon_r \approx 4.4$) is assumed. All six SMA connectors are on the left edge ($x = 10$ mm) at a pitch of 16 mm, arranged top to bottom as J4/J5 (channel 2), J2/J3 (channel 1) and J6/J7 (channel 3), so that the two electrodes of a channel are adjacent. USB-C, the external 5 V input, the UART, SWD and source-select headers are on the right edge. Figure 7 shows the copper of all four layers, rendered directly from the board file; Figure 8 shows the assembled top side.

Table 3: Layer assignment and principal copper statistics (from the KiCad board-statistics report and the geometry extraction of Section 6).

Layer	KiCad name	Function	Copper / routing content
L1	F.Cu	components, sense lines, SPI, local power	577 mm ² ; all six centre-pin lands
L2	In1.Cu (GND_PLANE)	ground reference plane	solid pour, 92.0 % of area, one polygon; 494 mm of tracks embedded
L3	In2.Cu (POWER_SIGNAL)	power distribution, signal escape	292 mm of tracks, no pour
L4	B.Cu	chassis pour under SMA area, SMA bottom lands	2620 mm ² , CHASSIS pour $x < 35$ mm

69 components (65 SMD, 4 THT); 275 SMD pads, 20 through-hole pads, 2 NPTH; 95 through vias (90×0.4 mm, 5×0.3 mm)
 minimum track width 0.20 mm; minimum measured clearance 0.1516 mm (rule 0.15 mm); minimum drill 0.30 mm

5.2 Sense-line routing

Each SMA centre land is followed by a straight 14.8 mm track on L1 to the 0Ω link; the continuation to the CDC port differs per electrode (Table 4). Channel 1 (PC0/PC1) is routed entirely on L1 over the ground plane. The other four electrodes change layer twice and run for 25–41 mm on L2, L3 or L4. For a 0.25 mm track over a 0.21 mm dielectric the microstrip capacitance is about 0.92 pF/cm, giving a board contribution of 3.4–7.5 pF per electrode against roughly 150 pF for 1.5 m of coaxial cable. The board therefore adds less than 5 % to the parasitic load that the external compensation must remove, and the absolute values are unimportant; what matters is that they are stable, which is the case for copper over a plane. Two aspects nevertheless deserve attention in a future revision: the 45 mm difference in path length between the two electrodes of channel 3 and the use of the ground-plane layer itself for two of the sense tracks (Section 7). The bottom-side chassis pour underlies the SMA area only; the centre-pin lands have no copper beneath them on L4, and a keep-out on L3 and L4 under the CDC package prevents pour copper directly below the sensitive port pins.

5.3 Power, ground and decoupling placement

Ground is carried by the L2 plane; the top layer has no ground pour, and every ground pad is connected to the plane by an individual via or a short track (36 ground vias in total). The

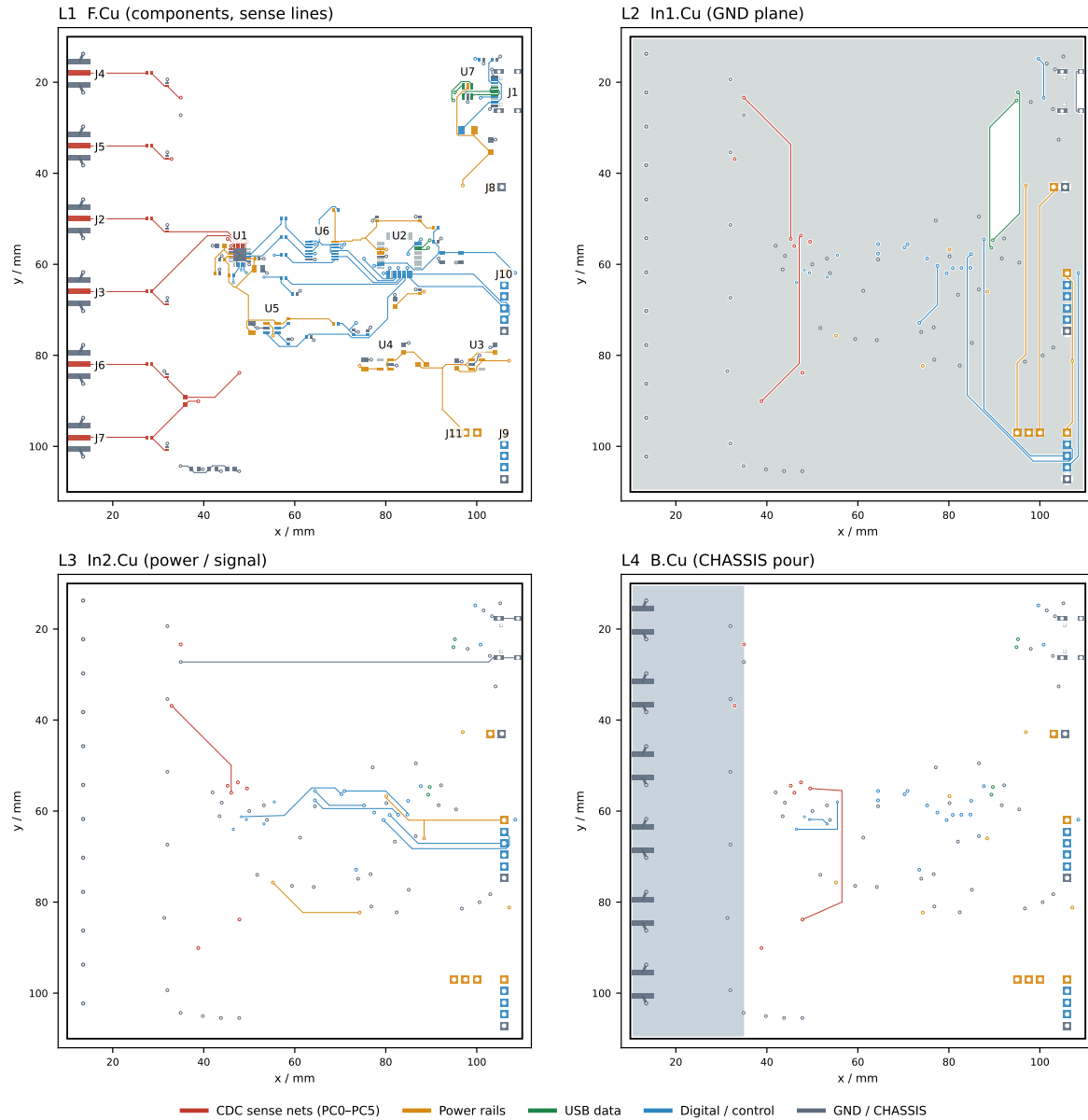


Figure 7: Copper of the four layers, rendered from the v1.5 board file. Sense nets are red, power orange, USB green, other digital signals blue, ground/chassis grey. Note the signal tracks embedded in the L2 ground plane (upper right panel) and the mixed-layer routing of the channel-2 and channel-3 sense lines.

Table 4: Routing of the six sense electrodes from the SMA land to the CDC port (extracted from the board file). The capacitance estimate uses 0.92 pF/cm irrespective of layer.

Port	Connector	Channel / electrode	Length / mm	Vias	Layers used
PC0	J2	1 inner	37.0	0	L1
PC1	J3	1 outer	44.6	0	L1
PC2	J4	2 inner	61.1	2	L1, L2 (35 mm)
PC3	J5	2 outer	45.7	2	L1, L3 (25 mm)
PC4	J6	3 inner	82.0	2	L1, L4 (41 mm)
PC5	J7	3 outer	75.7	2	L1, L2 (40 mm)

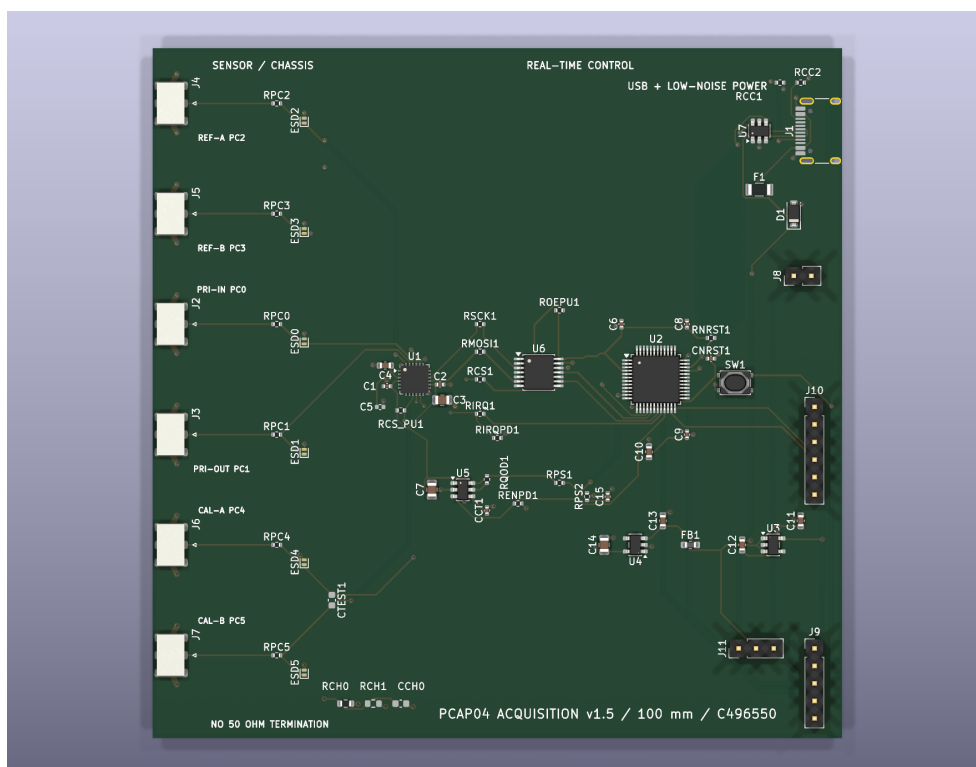


Figure 8: Rendered top view of the assembled board (KiCad 3D viewer; the SMA bodies are a simplified visual envelope only).

CDC decoupling is placed correctly: C1 and C2 are 1.7 mm and 1.3 mm from their VDD33 pins and C4 2.3 mm from VDD18. The microcontroller’s three decoupling capacitors, however, are 11–13 mm from pins 4–6, and the buffer’s 100 nF is 10 mm from pin 14; the ST datasheet asks for the 100 nF capacitors to be “as close as possible” to the chip [2]. The gated rail +3V3P is a 50 mm top-layer track of 0.20–0.25 mm width, adequate for the few milliamperes drawn by the CDC. The analogue regulator U4 and its bead sit 26 mm from the load switch and 40 mm from the CDC, on the opposite side of the board from the USB connector, consistent with the placement rule written in the schematic.

5.4 SMA edge-connector footprint

The BWSMA-KE-P001 is an offset-pin clamp connector for 1.6 mm boards. The local footprint places one 4.8×1.2 mm centre-pin land on L1 and four shield lands of the same size at ± 2.55 mm on L1 and L4, with the land centres 2.70 mm inboard of the routed edge so that 0.30 mm of copper-free margin remains. These values were taken from the vendor’s EasyEDA package (in 0.254 mm units) and are documented in the project’s footprint audit; the earlier v1.4 footprint, whose lands straddled the outline and required a negative copper-to-edge rule exception, is superseded. Because the connector’s mechanical fit cannot be proven by any electrical rule check, a physical trial fit on the first bare boards is mandatory before assembly.

5.5 Manufacturability

The design uses conservative geometry throughout: 0.2 mm minimum track, 0.15 mm clearance rule, 0.3 mm minimum via drill and 0.4 mm standard vias, all within the standard capability of low-cost four-layer services. The smallest measured clearance (0.1516 mm) is only 1.6 μ m above the rule, so the fabricator’s stated minimum should be confirmed as 0.15 mm or better. Gerber X2, Excellon PTH/NPTH files, an IPC-D-356 netlist and a JLCPCB-ordered upload archive (L2 =

GND_PLANE, L3 = POWER_SIGNAL) are part of the release; the online Gerber preview must be inspected for the layer order before ordering.

6 Verification

6.1 Method

The verification combined three independent sources of evidence.

1. **Native tool checks.** The release includes KiCad 10.0.5 reports: ERC with 0 errors and 0 warnings across the three sheets, DRC after copper refill with 0 violations and 0 unconnected items, and 0 schematic-to-board parity issues.
2. **Independent connectivity extraction.** A Python parser was written for the KiCad s-expression board format. It recovered all 69 footprints, 297 pads with their pin function and net, 507 track segments, 95 vias and 4 zones, and produced a complete net-to-pin table (Appendix B). Because the board file carries the pin functions copied from the schematic symbols, this table reflects the schematic as well.
3. **Datasheet cross-check.** For each integrated circuit the manufacturer’s current datasheet was obtained and every pin number was compared with the extracted pin function and net [1, 2, 3, 4, 5, 6, 7]. The comparison is exhaustive (109 IC pins plus the 20 receptacle contacts, Table 5). Application requirements (decoupling, enable thresholds, strapping, timing components) were checked in the same pass.

The layout analysis of Section 5 used the same extracted geometry.

6.2 Pin-level results

Table 5: Pin-assignment cross-check against manufacturer datasheets.

IC	Pins	Checked items	Result
U1 PCAP04 (QFN-24+EP)	25	PC0–PC5 on 22, 23, 24, 1, 20, 21; GND 2, 8, EP; VDD18 3; VDD33 4, 14; SSN 9, MISO 10, PG5 11, IIC_EN 13 = GND, MOSI 15, SCK 16; PT*, PCAUX, PG2–PG4 open	PASS
U2 STM32G0B1CBT6 (LQFP-48, GP)	48	all 48 pin names vs. DS13560 Fig. 5/Table 12 including VBAT 4, VREF+ 5, VDD/VDDA 6, VSS/VSSA 7, NRST 10, USB 33/34, SWD 35/36; no VDDIO2 pin in the GP variant	PASS
U3 TLV75533P (SOT-23-5)	5	IN 1, GND 2, EN 3, NC 4, OUT 5; C_{in} 1 μ F, C_{out} 4.7 μ F > 0.47 μ F eff.	PASS
U4 TPS7A2033P (SOT-23-5)	5	IN 1, GND 2, EN 3, NC 4, OUT 5; C_{in} 1 μ F, C_{out} 10 μ F $\geq$ 1 μ F	PASS
U5 TPS22918 (SOT-23-6)	6	VIN 1 = +3V3A, GND 2, ON 3, CT 4 (1 nF), QOD 5 (470 Ω to VOUT), VOUT 6 = +3V3P; $V_{IH,ON} \geq 1$ V	PASS
U6 SN74LV125A (TSSOP-14)	14	1OE/2OE/3OE/4OE on 1, 4, 10, 13; A→Y direction for buffers 1–3 host→CDC and buffer 4 CDC→host; VCC 14, GND 7	PASS
U7 USBLC6-2SC6 (SOT-23-6)	6	I/O1 on 1 and 6 (D+ in/out), I/O2 on 3 and 4 (D– in/out), GND 2, VBUS 5	PASS
J1 USB-C receptacle	16+4	A4/A9/B4/B9 VBUS joined; A6/B6 D+ and A7/B7 D– paired; CC1 A5, CC2 B5 each 5.1 k Ω to GND; SBU open; A1/A12/B1/B12 GND; shell to CHASSIS	PASS

All 109 IC pins and the USB receptacle are connected as the datasheets require. The U5 pin-order error that existed in revision v1.2 (and was corrected in v1.3) is confirmed to be absent: pins 1–6 are VIN, GND, ON, CT, QOD, VOUT in both schematic and board. No other pin-order transposition was found. Table 6 lists the application-level checks.

Table 6: Application-level checks derived from the datasheets and the schematic notes.

Check	Result
PCAP04 VDD33 buffer $\geq 10\ \mu\text{F}$ (C3+C7 = $20\ \mu\text{F}$), VDD18 $\geq 4.7\ \mu\text{F}$ (C4)	PASS
PCAP04 IIC_EN tied (not floating); SPI selected; VDD18 not connected to any supply	PASS
PCAP04 floating mode with internal reference: 3 sensors on PC0/1, PC2/3, PC4/5; C_REF_INT allowed (not differential)	PASS
Cable shields grounded at PCB end through single-point CHASSIS-GND bond	PASS
STM32 supplies: VDD/VDDA = VREF+ = VBAT = +3V3D; VDDUSB internal; HSI48 + CRS, no crystal	PASS
STM32 NRST: $100\ \text{nF} + 10\ \text{k}\Omega + \text{button}$, on SWD header; no capacitor on PA14/SWCLK	PASS
USB-C UFP: $R_d = 5.1\ \text{k}\Omega$ on both CC; VBUS fused (500 mA PTC) and TVS-clamped; ESD array in data path	PASS
LDO enables tied to inputs; bead isolation of analogue LDO input; sources never paralleled (jumper select)	PASS
Load switch default off ($100\ \text{k}\Omega$ pull-down), slew $1\ \text{nF}$, QOD active; rail monitor divider within ADC range	PASS
SPI buffer default tri-state ($100\ \text{k}\Omega$ pull-up on OE); CS pull-up referenced to gated rail; I_{off} outputs	PASS
STM32 $100\ \text{nF}$ decoupling “as close as possible”	NOTE (11–13 mm)
LV125A unused/idle inputs must not float (A4 = PCAP_MISO has no bias resistor)	NOTE
Schematic requirement “L2 continuous GND, do not cut the ground plane”	NOTE (494 mm of tracks in L2)
Schematic requirement “USB D+/D– $90\ \Omega$ differential, equal length”	NOTE (L2 routing, 5 mm skew)

7 Findings and recommendations

Table 7 summarises the observations. None is a wiring error and none blocks fabrication of the bare board; items F1–F4 are recommended for a v1.6 layout revision, the remainder are component or firmware choices that can be applied at assembly or bring-up.

7.1 Discussion of the principal findings

Ground-plane integrity (F1, F2). A four-layer board with a dedicated ground layer achieves its low-impedance return path only if that layer is uninterrupted beneath the signals that reference it. In v1.5 the router placed about half a metre of track in L2, including the USB pair and both SWD lines, which run from the right-hand connectors to the microcontroller through channels cut in the plane, and two of the six sense lines. For the sense lines the consequence is not a change of the measured capacitance – the compensation sequence removes any static parasitic – but a loss of the shielding that the plane provides against interference coupling into a high-impedance node during the discharge interval, and an asymmetry between the two electrodes of a channel. Since L3 carries only 292 mm of tracks and no pour, all L2 tracks can be moved there without

Table 7: Findings. Severity: **FIX** = should be corrected in the next revision; **NOTE** = advisable improvement or firmware/assembly action. Pin-out (109 IC pins, U5 order, USB receptacle) and rule checks (ERC 0/0, DRC 0, parity 0) are **PASS**.

#	Area	Observation and recommendation	Severity
F1	Layout, L2	494 mm of signal and power tracks (USB D±, SWD, NRST, +5V_USB, +5V_EXT, +3V3D, two sense lines) are routed <i>in</i> the ground-plane layer, contrary to the schematic’s own requirement. The plane remains one polygon (92% fill) but its return-path continuity under the SPI and USB lines is interrupted by channels. Move these tracks to L3 (which carries no pour and only 292 mm of tracks) and keep L2 solid.	FIX
F2	Layout, sense	PC2 and PC5 run 35–40 mm on the plane layer, PC3 on L3 and PC4 on L4 through the chassis pour; channel 3 has a 45 mm length imbalance between its electrodes. Route all six sense lines on L1 over the solid plane with no vias, and match the two electrodes of each channel to within a few millimetres so that $C_{pA} \approx C_{pB}$ and the compensation residual is minimised.	FIX
F3	Layout, decoupling	C8, C9, C10 are 11–13 mm from the STM32 supply pins and C6 10 mm from the buffer’s VCC. Place the 100 nF parts within 2 mm of pins 5/6 and 14 respectively.	FIX
F4	Layout, USB	D+/D– are routed as two single-ended tracks on L2 with 5 mm skew, not as a 90 Ω pair. Irrelevant for full-speed signalling over 40 mm, but the schematic asks for a controlled pair; route on L1 over the plane as a coupled pair for consistency.	FIX
F5	SPI buffer	Buffer input A4 (PCAP_MISO) is driven only by the CDC, which is unpowered for part of the operating cycle and may release MISO when deselected; the LV125A datasheet requires inputs never to float. Add a 100 kΩ pull-down from PCAP_MISO to GND.	NOTE
F6	Interrupt	RIRQPD1 (10 kΩ to GND) draws 0.33 mA from the gated rail whenever INTN idles high (INTN is active low), and with the CDC unpowered the line reads as “asserted”. Increase to 100 kΩ; mask the EXTI in firmware until the rail is confirmed up.	NOTE
F7	Silkscreen	The connector labels (REF-A PC2, REF-B PC3, PRI-IN PC0, PRI-OUT PC1, CAL-A PC4, CAL-B PC5) describe an older allocation and contradict the schematic (LEVEL1–3 inner/outer). Update the silkscreen to the schematic names to avoid mis-cabling.	NOTE
F8	Temperature port	PTOUT/PT0REF/PT1 are open, so the CDC’s temperature (RDC) measurement is unavailable for drift correction. If temperature compensation is wanted later, add a 10 nF C0G position at PTOUT and a reference-resistor position at PT0REF (both DNP).	NOTE
F9	Reference	With three floating sensors the internal reference (≤ 33 pF) must be used; absolute accuracy then depends on its tolerance and drift. Plan a gain calibration against the CTEST1 position (or an external 0.1% capacitor) and record CDC_GAIN_CORR; keep the PC0/PC1-as-reference option in mind if two channels suffice.	NOTE
F10	Manufacturing	Minimum measured clearance is 0.1516 mm, 1.6 μm above the 0.15 mm rule; confirm the fabricator’s four-layer capability, and verify L2/L3 order in the Gerber preview. BOM entries other than J2–J7 have no verified supplier part numbers; placement rotations are not yet checked in the assembler’s preview.	NOTE
F11	Mechanical	SMA land geometry follows the vendor package, but clamp fit on 1.6 mm and the 0.30 mm copper-to-edge margin cannot be verified electrically. Trial-fit all six connectors on the first bare boards before ordering assembly.	NOTE

congestion, after which L2 becomes the solid plane the schematic asks for. The sense lines are best kept on L1 directly over the plane, without vias.

Decoupling placement (F3). The electrical function is not affected on a board of this size, but the 11–13 mm loop between the STM32 and its capacitors increases the supply-loop inductance and radiated emission of the 64 MHz core. The fix is a placement change only.

Floating buffer input (F5). When the CDC rail is off, the `PCAP_MISO` node is held only by the leakage paths of the unpowered CDC and sits at an undefined level below the input threshold of the buffer; a CMOS input in that state can draw excess supply current and is explicitly disallowed by the buffer datasheet. A single 100 k Ω resistor resolves this and also defines the line while the CDC is deselected.

Interrupt polarity (F6). The design treats the CDC interrupt as active high in its naming (`PCAP_IRQ` with a pull-down), whereas the PCAP04 `INTN` output is active low and is released high by the SPI transaction that reads the results. The hardware works either way, but the firmware must configure the EXTI for a falling edge and ignore the line while the rail is down. The pull-down value is a minor power and loading issue on the low-noise rail.

8 Bring-up and operating procedure

The schematic prescribes a safety sequence that the firmware must implement; it follows directly from the interlock circuit of Figure 5 and is illustrated in Figure 9.

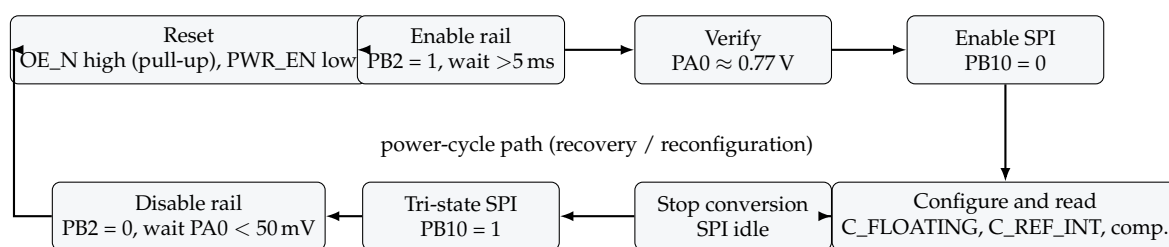


Figure 9: Firmware sequence implied by the hardware interlock. The SPI buffers are enabled only between rail-up verification and rail-down, so that the unpowered CDC is never driven.

For the first board the following order is recommended. (1) Before power, verify the U5 pin order and the six SMA fits, and measure +3V3P to GND for shorts. (2) Power through a current-limited supply on J8 with J11 in position 2–3; check +3V3D and +3V3A at 3.3 V and +3V3P at 0 V. (3) Programme the microcontroller over SWD (J9) or over USB DFU (a blank device boots into the ROM bootloader). (4) Assert `PCAP_PWR_EN` from firmware and confirm the rise time (about 2 ms) and the ADC reading; de-assert and confirm the decay to below 50 mV within 100 ms. (5) Enable the buffers and read the PCAP04 identification; write the configuration with `C_FLOATING=1`, `C_DIFFERENTIAL=0`, `C_REF_INT=1` with `C_REF_SEL=31`, `C_COMP_INT=1`, `C_COMP_EXT=1`, `C_PORT_EN=0x3F`, `RDCHG_INT_SEL0/1=30 k $\Omega$` and `PG5_INTN_EN=1`. (6) With a 100 pF C0G capacitor in the CTEST1 position and nothing on the SMA inputs, determine the gain-correction factor and the noise floor at the intended data rate. (7) Connect the cables and electrodes, verify that the external compensation removes the cable capacitance (result independent of cable length to first order), and record the final register set in the firmware.

9 Conclusion

Revision v1.5 of the PCAP04 acquisition board is electrically correct: an exhaustive, tool-independent cross-check of all integrated-circuit pins against the current datasheets found no wiring error, the datasheet application requirements for decoupling, strapping, enable levels and timing components are met, and the native rule checks are clean. The architecture – floating sensor connection with full parasitic compensation, single-point shield grounding, a separately regulated and switchable analogue rail, and a tri-state isolated SPI bus – is well matched to the task of reading cable-connected capacitive level sensors. The weaknesses identified are confined to the layout: signal routing inside the ground-plane layer, mixed-layer and unbalanced sense-line routing, and distant microcontroller decoupling. These do not prevent the bare board from being fabricated and used for prototyping, but they should be corrected, together with the two small component additions (a MISO pull-down and a weaker interrupt pull-down) and the silkscreen naming, in revision v1.6 before a larger batch is ordered.

References

- [1] ScioSense B.V., *PCAP04 Capacitance-to-Digital Converter Datasheet*, SC-001050-DS-6, Sept. 2023.
- [2] STMicroelectronics, *STM32G0B1xB/xC/xE Arm Cortex-M0+ 32-bit MCU Datasheet*, DS13560 Rev. 6.
- [3] Texas Instruments, *TLV755P 500-mA, Low-IQ, Small-Size, Low-Dropout Regulator*, SBVS319.
- [4] Texas Instruments, *TPS7A20 300-mA, Ultra-Low-Noise, Low-IQ, Low-Dropout Voltage Regulator*, SBVS351.
- [5] Texas Instruments, *TPS22918 5.5-V, 2-A, 52-mΩ On-Resistance Load Switch*, SLVSD07.
- [6] Texas Instruments, *SN74LV125A Quadruple Bus Buffer Gate with 3-State Outputs*, SCLS380.
- [7] STMicroelectronics, *USBLC6-2 Very Low Capacitance ESD Protection*, DS5238.
- [8] KiCad Developers, *KiCad EDA 10.0.5*, project files PCAP04_Acquisition_3CH_v1.5_100mm_C496550, release notes 2026-08-25.

A Bill of materials

Table 8: Bill of materials of revision v1.5 (from the KiCad export). DNP = not fitted on the first build.

References	Value	Footprint	Qty	Fit
U1	PCAP04-AQFM-24	QFN-24 4×4, EP 2.7×2.7	1	
U2	STM32G0B1CBT6	LQFP-48 7×7	1	
U3	TLV75533PDBVR	SOT-23-5	1	
U4	TPS7A2033PDBVR	SOT-23-5	1	
U5	TPS22918DBVR	SOT-23-6	1	
U6	SN74LV125APWR	TSSOP-14	1	
U7	USBLC6-2SC6	SOT-23-6	1	
J1	USB-C 2.0 receptacle, HRO TYPE-C-31-M-12	16-pin hybrid	1	
J2–J7	BWSMA-KE-P001 (LCSC C496550)	edge-mount SMA, local footprint	6	
J8	EXT_5V_LOW_NOISE	1×2 header 2.54	1	
J9	SWD	1×5 header 2.54	1	
J10	UART_HOST	1×6 header 2.54	1	
J11	5V_SOURCE_SELECT	1×3 header 2.54	1	
SW1	RESET	SW_SPST_PTS810	1	
F1	PTC 500 mA	1206	1	
D1	SMF5.0A	SOD-123F	1	
FB1	600 Ω @ 100 MHz	0603	1	
ESD0–ESD5	ESD diode	SOD-882	6	DNP
C1, C2, C6, C8, C9, CNRST1	100 nF	0402	6	
C3, C7, C14	10 μ F	0805	3	
C4, C10, C12	4.7 μ F	0603	3	
C5	100 nF	0402	1	DNP
C11, C13	1 μ F	0603	2	
C15	10 nF C0G	0402	1	
CCT1	1 nF	0402	1	
CCH0	1 nF	0603	1	DNP
CTEST1	100 pF C0G 0.1 %	0603	1	DNP
RPC0–RPC5	0 Ω	0402	6	
RCH0	0 Ω	0603	1	
RCH1	1 M Ω	0603	1	DNP
RCC1, RCC2	5.1 k Ω 1 %	0402	2	
RCS1, RMOSI1, RSCK1	33 Ω	0402	3	
RCS_PU1, RENPD1, ROEPU1, RPS2	100 k Ω	0402	4	
RIRQ1	1 k Ω	0402	1	
RIRQPD1, RNRST1	10 k Ω	0402	2	
RPS1	330 k Ω	0402	1	
RQOD1	470 Ω	0402	1	

B Net list summary

The following table lists every named net with its connections as extracted from the board file (pad functions in parentheses); locally named two-pin nets are given in abbreviated form.

Table 9: Net connectivity of revision v1.5.

Net	Connections
+3V3A	C14, U4.5 (OUT), U5.1 (VIN)
+3V3D	C6, C8, C9, C10, C12, J9.1, J10.1, RNRST1, ROEPU1, U2.4/5/6 (VBAT, VREF+, VDD/VDDA), U3.5 (OUT), U6.14 (VCC)
+3V3P	C1, C2, C3, C7, RCS_PU1, RPS1, RQOD1, U1.4 (VDD33A), U1.14 (VDD33B), U5.6 (VOUT)
+5V_AIN	C13, FB1.2, U4.1 (IN), U4.3 (EN)
+5V_EXT	J8.1, J11.3
+5V_IN	C11, FB1.1, J11.2, U3.1 (IN), U3.3 (EN)
+5V_USB	D1 (K), F1.2, J11.1, U7.5 (VBUS)
VDD18_LOCAL	C4, C5 (DNP), U1.3 (VDD18)
CHASSIS	CCH0, ESD0–5 (A), J1 shell, J2–J7 shield lands, RCH0, RCH1
GND	all ground pins and capacitor returns: U1.2/8/13/EP, U2.7, U3.2, U4.2, U5.2, U6.7, U7.2, J1 A1/A12/B1/B12, J8.2, J9.5, J10.6, D1 (A), SW1.2, RCC1/2, RCH0/1, RENPD1, RIRQPD1, RPS2, C1–C15, CCH0, CCT1, CNRST1
NRST	CNRST1, J9.4, RNRST1, SW1.1, U2.10
PC0 ... PC5	J2.1–RPC0–U1.22 (PC0); J3.1–RPC1–U1.23 (PC1); J4.1–RPC2–U1.24 (PC2); J5.1–RPC3–U1.1 (PC3); J6.1–RPC4–U1.20 (PC4), CTEST1.1; J7.1–RPC5–U1.21 (PC5), CTEST1.2; each port node also to ESD _n (K)
SPI host side	U2.16 (PA5)–U6.2 (A1); U2.18 (PA7)–U6.5 (A2); U2.19 (PB0)–U6.9 (A3); U2.17 (PA6)–U6.11 (Y4)
SPI CDC side	U6.3 (Y1)–RSCK1–U1.16 (SCK); U6.6 (Y2)–RMOSI1–U1.15 (MOSI); U6.8 (Y3)–RCS1–U1.9 (SSN), RCS_PU1; U1.10 (MISO)–U6.12 (A4)
PCAP_SPI_OE_N	ROEPU1, U2.22 (PB10), U6.1/4/10/13 (1OE–4OE)
PCAP_IRQ	U1.11 (PG5), RIRQPD1, RIRQ1 → U2.20 (PB1)
PCAP_PWR_EN	RENPD1, U2.21 (PB2), U5.3 (ON)
PCAP_PWR_SENSE	C15, RPS1, RPS2, U2.11 (PA0)
U5 timing	U5.4 (CT)–CCT1; U5.5 (QOD)–RQOD1
USB	J1 A6/B6–U7.1 (D+ in); J1 A7/B7–U7.3 (D– in); U7.6–U2.34 (PA12, USB_DP); U7.4–U2.33 (PA11, USB_DM); J1 A4/A9/B4/B9–F1.1; J1 A5–RCC1; J1 B5–RCC2
Headers	J9: +3V3D, SWDIO (PA13), SWCLK/BOOT0 (PA14), NRST, GND; J10: +3V3D, UART_TX (PA2), UART_RX (PA3), HOST_READY (PB11), MEAS_BUSY (PB12), GND